

Bauchi State University, Gadau

Faculty of Science – Department of Biological Sciences

Course Code: BIO1201 – General Biology

Academic Session: 2023/2024 – First Semester Examination

Time Allowed: 1 Hour 30 Minutes

Instruction: ANSWER ALL QUESTIONS

Questions 1–60

1. Which of the following does not play a role in cytokinesis?

- (a) Cell membrane
- (b) Cell plate
- (c) Golgi apparatus ☒
- (d) Cell wall

2. The process that produces two identical daughter cells is called

- (a) Mitosis ☒
- (b) Meiosis
- (c) Cytokinesis
- (d) Telophase

3. The cell in which the nucleus is often called

- (a) Prokaryotic
- (b) Eukaryotic ☒
- (c) Unicellular
- (d) Multicellular

4. The following are products of Meiosis except

- (a) Gamete production
- (b) $4n$ found in haploid chromosome ☒
- (c) 2 stages division occurs
- (d) 4 daughter cells

5. The first stage of cell division is called

- (a) Growth phase ☒
- (b) Cell cycle
- (c) Life cycle
- (d) Cytokinesis

6. Which type of cells does Mitosis occur?

- (a) All living cells
- (b) Somatic cells ☒
- (c) Reproductive cells

(d) Nerve cells

7. RNA simply stands for

- (a) Ribonucleic Acid ✓
- (b) Ribonucliec Acid
- (c) Ribonucleic Acid
- (d) Ribosensic Acid

8. When two genes are identical (RR and rr), they are called

- (a) Dominancy
- (b) Homozygous ✓
- (c) Heterozygous
- (d) Monozygous

9. The Nucleus contains all except

- (a) Membrane ✓
- (b) Nucleolus
- (c) Chromosomes
- (d) All of the above

10. The differences between prophase of Meiosis and Mitosis

- (a) No Synapsis and crossing over
- (b) Synapsis and crossing over ✓
- (c) Chromosomes
- (d) All of the above

11. The crossover of Synapsis and crossing over occurs in

- (a) 23
- (b) 46
- (c) 23
- (d) 46 ✓

12. The total number of chromosomes in human cell is

- (a) 23
- (b) 46 ✓
- (c) 23
- (d) 46

13. The organelle that maintains the shape of a cell is called

- (a) Lysosomes
- (b) Mitochondria
- (c) Golgi apparatus
- (d) Ribosome ✓

14. The organelle that maintains the shape of a cell is called

- (a) Lysosomes
- (b) Mitochondria
- (c) Golgi apparatus
- (d) Ribosome ✓

15. The organelle that helps in the production of protein is called

- (a) Lysosomes
- (b) Mitochondria
- (c) Golgi apparatus
- (d) Ribosome ✓

16. By gene influence

- (c) Chromosomal mutation ✓
- (d) DNA Splicing

17. The process that results in changes in the hereditary material is called

- (a) Transformation
- (b) Metamorphosis
- (c) Replication
- (d) Mutation ✓

18. The stage of cell cycle in which DNA is synthesized is

- (a) Gap 1
- (b) Mitotic phase
- (c) Gap 2
- (d) S-phase ✓

19. The cell in which the true nuclei is sealed

- (a) Prokaryotic
- (b) Eukaryotic ✓
- (c) Unicellular
- (d) Multicellular

20. Robert Hooke postulated that all living things are made up of

- (a) Cells ✓
- (b) Organelles
- (c) Nucleus
- (d) Chromosomes

21. The cell organelle that occurs only in animals is

- (a) Cell membrane
- (b) Cell wall
- (c) Cell organelles
- (d) Centrioles ✓

22. The failure of chromosomes to separate during cell division is called

- (a) Cell membrane
- (b) Cell organelles
- (c) Nucleus
- (d) Independent assortment ☒

23. The organelle where the cell's used parts are contained

- (a) Cell membrane
- (b) Cell wall
- (c) Lysosomes ☒
- (d) Mitochondria

24. Animal growth and development can be separated into

- (a) Sexual and Asexual
- (b) Prenatal and postnatal ☒
- (c) Mitosis and Meiosis
- (d) Growth and development

25. All the following are importance of cellular division except

- (a) Growth
- (b) Development
- (c) Energy ☒
- (d) Repair

26. RNA can be categorized into the following except

- (a) Messenger RNA
- (b) Transfer RNA
- (c) Ribosomal RNA
- (d) Ribosensic RNA ☒

27. The following are characteristics of Mitosis except

- (a) Occurs in somatic cells
- (b) Ends in haploid ☒
- (c) Reproductive cells
- (d) One stage division occurs

28. It occurs in somatic cells

- (a) All living cells
- (b) All living cells
- (c) Somatic cells ☒
- (d) Reproductive cell

29. Which of the following is NOT a characteristic of living things?

- (a) Organization
- (b) Homeostasis

(c) Inorganic composition ☒

(d) Growth and development

30. What is the basic unit of life?

(a) Atom

(b) Cell ☒

(c) Molecule

(d) Organ

31. The process by which living organisms maintain internal stability

(a) Metabolism

(b) Reproduction

(c) Homeostasis ☒

(d) Adaptation

32. Which of the following is an example of a stimulus that living organisms can respond to?

(a) Gravity

(b) None of the above

(c) Light

(d) All of the above ☒

33. What is the term for acquiring and using energy by living organisms?

(a) Homeostasis

(b) Metabolism ☒

(c) Growth

(d) Reproduction

34. Which characteristic involves increase in size or number of cells?

(a) Growth ☒

(b) Nutrition

(c) Adaptation

(d) Reproduction

35. Ability to change and adapt to environment over time is called

(a) Reproduction

(b) Adaptation ☒

(c) Homeostasis

(d) Growth

36. Role of DNA in living organisms

(a) Internal stability

(b) Provides energy

(c) Carries genetic info ☒

(d) Disrupts reproduction

37. Example of ecological relationship

- (a) Competition between species ☒
- (b) Photosynthesis
- (c) Cell division
- (d) Homeostasis

38. Characteristic of living things: ability to learn and communicate

- (a) Mortality
- (b) Adaptation
- (c) Consciousness ☒
- (d) Growth and development

39. Passing on genetic info from parents to offspring

- (a) Homeostasis
- (b) Evolution
- (c) Reproduction
- (d) Heredity ☒

40. Metabolism is the process of acquiring and converting energy

☒ TRUE

41. Characteristic involving complexity and new abilities

- (a) Growth
- (b) Development ☒
- (c) Adaptation
- (d) Reproduction

42. Irritability is not a characteristic of living things

☒ FALSE

43. Responsible for internal stability

- (a) Homeostasis ☒
- (b) Adaptation
- (c) Reproduction
- (d) Growth

44. Respiration is the process of acquiring energy

☒ TRUE

45. Process of increasing in size or number of cells

- (a) Development
- (b) Growth ☒
- (c) Reproduction
- (d) Adaptation

46. Stimulus that organisms respond to

- (a) Wood
- (b) Sound ✓
- (c) Gravity
- (d) Inanimate matter

47. Responsible for passing genetic info

- (a) RNA
- (b) DNA ✓
- (c) Protein
- (d) Carbohydrate

48. Process of genetic changes enhancing survival

- (a) Adaptation
- (b) Growth
- (c) Reproduction
- (d) Evolution ✓

49. Responsible for continuity of species

- (a) Growth
- (b) Reproduction ✓
- (c) Development
- (d) Metabolism

50. Meaning of DNA

- (a) Deoxyribonucleic Acid ✓
- (b) Dyribonucleic Acid
- (c) Deoxybribonucleic Acid
- (d) Deoxyribonucliec Acid

51. Responsible for evolution of species

- (a) Adaptation
- (b) Reproduction
- (c) Growth
- (d) Natural selection ✓

52. Genetic material in organisms

- (a) RNA
- (b) Proteins
- (c) DNA ✓
- (d) Carbohydrates

53. Ability to use materials for cellular activities

- (a) Reproduction
- (b) Respiration ✓

- (c) Homeostasis
- (d) Adaptation

54. Interconnected relationship in ecosystem

- (a) Homeostasis
- (b) Adaptation
- (c) Metabolism
- (d) Ecological relationships ☒

55. Ability to adapt to new conditions

- (a) Homeostasis
- (b) Growth
- (c) Reproduction
- (d) Adaptation ☒

56. Responsible for perception and interaction

- (a) Homeostasis
- (b) Adaptation
- (c) Growth
- (d) Consciousness ☒

57. Finite lifespan of organisms

- (a) Mortality ☒
- (b) Immortality
- (c) Adaptation
- (d) Reproduction

Q59. Which of the following describes the finite lifespan of living organisms?

- (a) Mortality ☒
- (b) Immortality
- (c) Adaptation
- (d) Reproduction